FIITJEE

ALL INDIA TEST SERIES

FULL TEST – III

JEE (Advanced)-2020

PAPER – 2

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. B

Sol. $\Delta\phi = \frac{2\pi}{\lambda}\Delta x = (2n+1)\frac{\pi}{2}$

2. A

Sol. $v = \sqrt{\frac{T}{\mu}}$ $\qquad V' = \frac{6V}{5}$

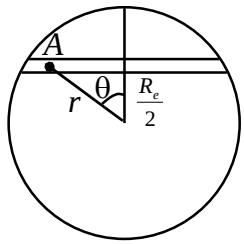
$V' = \sqrt{\frac{T'}{\mu}}$ $\qquad T' = \frac{36}{25}T$

% T = 44 %

3. D

Sol. Let mass of the earth is M_e and mass of object is m.

Force on object at $A = \left(\frac{GM_e m}{R^3}r\right)$ towards centre.



Force on object along the tunnel $= \left(\frac{GM_e m}{R^3}r\right)\frac{\sqrt{r^2 - \frac{R^2}{4}}}{r}$

Acceleration of object along the tunnel $= \left(\frac{GM_e}{R^3}\right)\sqrt{r^2 - \frac{R^2}{4}}$

4. A

Sol. $\Delta Pm = 2\Delta P_0 \cos kx$ (assuming closed end as origin)

At point Q, $x = L - \frac{7L}{9} = \frac{2L}{9}$

$\Delta Pm = 2\Delta P_0 \cos\left(\frac{2\pi}{\lambda} \times \frac{2L}{9}\right) = \Delta P_0$

$\therefore$ Required ratio = 1 : 2

5. B

Sol. $d = \frac{mv_0}{qB} + \frac{2mv_0}{2qB}$

6. C

Sol. $\frac{hc}{\lambda} = R\left(\frac{1}{1^2} - \frac{1}{2^2}\right) = \frac{3}{4}R$ …(i)

$\frac{hc}{\lambda'} = R\left(\frac{1}{1^2} - \frac{1}{3^2}\right) = \frac{8}{9}R$ …(ii)

$\frac{\lambda'}{\lambda} = \frac{3}{4} \times \frac{9}{8}$, $\lambda' = \frac{27}{32}\lambda$

7. C

Sol. $\mu = \frac{3}{2}$; $V = 8$; $m = \frac{1}{3}$

$1 + m = \frac{V}{f} = \frac{V}{2R}$

8. BD

Sol. $R = \frac{l}{|\alpha_B - \alpha_C|\Delta t}$

9. AD

Sol. $\frac{d(emf)}{dt} = Bu\frac{d\ell}{dt}$ = constant

$i = \frac{Bu}{\lambda}$ = constant

10. AC

Sol. All the mass defect will convert into KE of α-particle.

11. BCD

Sol. $TP^{-2/5}$ = constant

$\Rightarrow$ $PV^{5/3}$ = constant or $P^{3/5}V = k$

This an adiabatic process.

$\Rightarrow$ $\Delta H = 0$

Work done = $-\Delta U = 6RT$

$$\int VdP = \int \frac{k\,dP}{P^{3/5}} = \frac{nR\Delta T}{(5/2)} = 10\ RT$$

12. AC

Sol. Let I be the power of source S_1 and S_2.

Initially $I_o = 4I$, finally $I_{max} = \left(\sqrt{I} + \sqrt{0.36I}\right)^2 = 2.56I$

and $I_{min} = \left(\sqrt{I} - \sqrt{0.36I}\right)^2 = 0.16I$

13. ABCD

Sol. $y = 2a \cos(\omega t + \phi) \cos(Kx)$ comparison with given equation gives

$$K = \frac{2\pi}{\lambda} = 10\pi \Rightarrow \lambda = \frac{1}{5}m$$

$\omega = 2\pi f = 50\pi \Rightarrow f = 25$ Hz

$\therefore v = 5\ ms^{-1}$

at $x = 0.15$ m

$\cos(10\pi \times 0.15) = \cos(1.5\pi) = -1$ for all t

at $x = 0.3$

$\cos(10\pi \times 0.3) = \cos 3\pi = -0$ for all t

14. AD

Sol. At highest point, linear momentum

$2mv = mv_1 - mv$

$v_1 = 3v$

At highest point, velocity of other part after explosion = 3v

Centre of mass of particles will be at a distance R from initial point on ground.

15. A

Sol. Work function of the metal (ϕ) = $\frac{hc}{\lambda_{green}} = \frac{12408}{4963} = 2.5\ eV$

No. of photon emitted from the power source per unit time = $\frac{40}{2.5 \times 1.6 \times 10^{-19}} = 10^{20}$ photons

No. of photons incident on the metallic surface per unit time = $\frac{10^{20} \times \pi \times (1 \times 10^{-2})^2}{4\pi(1)^2} = 2.5 \times 10^{15}$ photons

No. of photoelectrons coming out from the metal surface per unit time

$= \frac{2.5 \times 10^{15}}{10^6} = 2.5 \times 10^9$ photoelectrons

16. B

Sol. The emission of photoelectron will stop when $\frac{hc}{\lambda_{violet}} = \varphi + eV$, where V is the potential of sphere.

$\frac{12408}{4136} = 2.5 + eV \quad ; \quad V = 0.5\,V$

17. C

Sol. $T_B = \frac{P_B}{P_A} T_A = 2T_0$

$(2T_0)V_0^{\frac{13}{9}-1} = T_C \cdot (2V_0)^{\frac{13}{9}-1} \Rightarrow T_C = 2^{5/9} T_0$

18. B

Sol. Heat is absorbed during the process AB and BC

$Q_{AB} = \frac{3}{2}R(T_B - T_A) = \frac{3}{2}RT_0$

$Q_{BC} = \left(\frac{3}{2}R - \frac{9}{4}R\right)(T_C - T_B) = \frac{3}{4}R \times (2 - 2^{5/9})T_0$

$Q_{abs} = \frac{3}{4}(4 - 2^{5/9})RT_0 \cong \frac{15}{8}RT_0 [2^{5/9} \cong 1.5]$

$Q_{rej} = \frac{3}{8}(2 + 2^{5/9})RT_0 \cong \frac{21}{16}RT_0$

$Q_{CD} = \frac{3}{2}R(T_D - T_C)$

$Q_{DA} = \left(\frac{3}{2}R - \frac{9}{4}R\right)(T_A - T_D)$

$W = \frac{9}{16}RT_0 \quad ; \quad \eta = \frac{W}{Q_{abs}} = 30\%$

Chemistry

PART – II

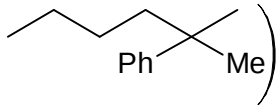
SECTION – A
(One Options Correct Type)

This section contains **07 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

19. C

Sol. $RCOOH \xrightarrow{SOCl_2} RCOCl \xrightarrow{CH_2N_2} RCOCHN_2 \xrightarrow{Ag_2O} RCH=C=O$

$RCH=C=O \xrightarrow{EtOH} RCH_2-C(=O)-OEt$

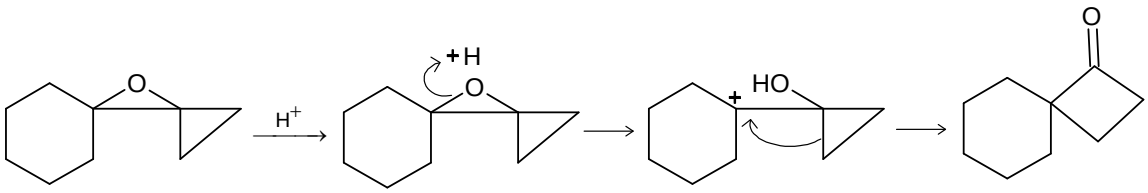
(R = 

)

20. D

Sol. Fact based.

21. C

Sol. 

22. C

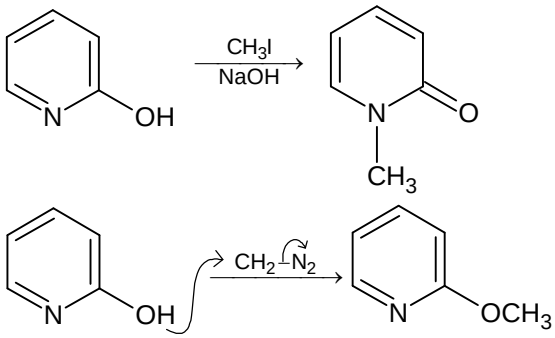
Sol. $Au + 4H^+ + NO_3^- + 4Cl^- \longrightarrow AuCl_4^- + NO + 2H_2O$

$3Pt + 16H^+ + 4NO_3^- + 18Cl^- \longrightarrow 3PtCl_6^{2-} + 4NO + 8H_2O$

23. A

Sol. Fact based.

24. C

Sol. 

25. A

Sol. For N_2

$$KK\ \sigma 2s^2 < \sigma^* 2s^2 < \sigma 2P_x^2 < \begin{matrix}\pi 2P_y^2 \\ \pi 2P_z^2\end{matrix} < \sigma 2P_x^2$$

For O_2^+

$$KK\ \sigma 2s^2 < \sigma^* 2s^2 < \sigma 2P_x^2 < \begin{matrix}\pi^* 2P_y^2 \\ \pi^* 2P_z^2\end{matrix} < \begin{matrix}\pi 2P_y^1 \\ \pi 2P_z\end{matrix} < \sigma^* 2P_z$$

26. ACD

Sol. (B) is incorrect, correct order is $Cl_2 > Br_2 > F_2 > I_2$.

27. AB

Sol. Fact based.

28. ABD

Sol. (A) $2H_2O \longrightarrow O_2 + 4H^+ + 4e^- \ (\text{at anode})$

(B) Aq. $AgNO_3$ at anode $4OH^- \longrightarrow 2H_2O + O_2 + 4e^-$

(D) Aq. NaOH at anode $4OH^- \longrightarrow 2H_2O + O_2 + 4e^-$

29. ACD

Sol. $K_2Cr_2O_7/H^+$ acts as oxidant not in basic medium, rest all acts as oxidizing agent in acidic as well as alkaline medium.

30. ABC

Sol. (A) Stephan's reduction, (B) Etard's reaction, (C) Gattermann Koch synthesis

31. ACD

Sol. $\underset{1-\alpha}{2X} \rightleftharpoons \underset{\frac{\alpha}{2}}{X_2} \qquad i = 1 - \frac{\alpha}{2}$

$$\frac{(\Delta t_b)_{0.65}}{K_b t} = i = 1 - \frac{\alpha}{2}$$

$$\frac{\alpha}{2} = 1 - \frac{\Delta t_b}{K_b t} = \frac{K_b t - \Delta t_b}{K_b t}$$

$$(A) \quad \alpha = \frac{2(K_b t - \Delta t_b)}{K_b t}$$

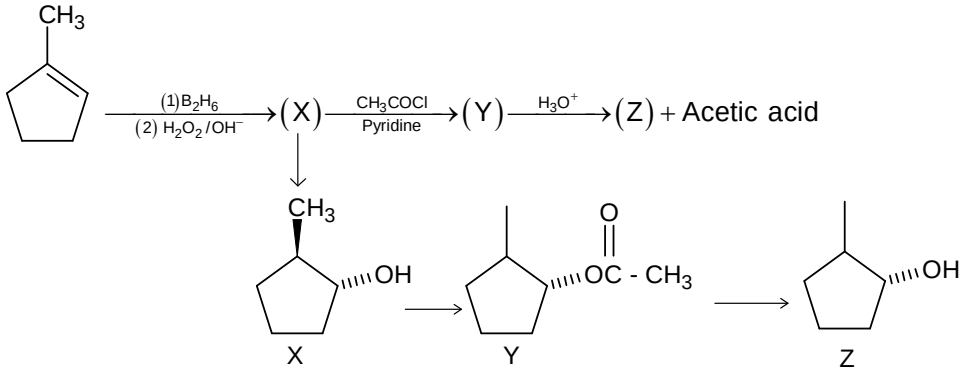
and as $i < 1$

$$\frac{M_C}{M_o} < 1 \text{ so } M_C < M_o$$

$$\frac{(C.P)_{obs}}{(C.P)_{cal}} < 1 \qquad .....(C)$$

32. AD

Sol.



33. D

Sol. $MnO_4^- + OH^- \longrightarrow MnO_2$

34. D

Sol. $KMnO_4$ is purple coloured due to charge transfer.

35. B

Sol. For 50% ionization(In^-) = [HIn]

$\therefore$ pH = PK_{In}

Thus $PK_{In} = 10^{-5}$ (as pH = 5)

36. A

Sol. At midway pH = PK_{In}

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **07 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

37. D

Sol. $f(2) < 0$

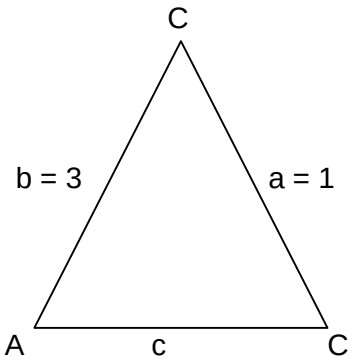
$\Rightarrow (a-6)(2a-3) < 0 \Rightarrow a \in \left(\frac{3}{2}, 6\right)$

38. A

Sol. By Cosine law, $\cos 60^o = \frac{1^2 + 3^2 - c^2}{2.1.3}$

$\Rightarrow c = \sqrt{7}$

Now, $\frac{b}{\sin B} = \frac{c}{\sin C}$ (By sine law)



$\Rightarrow \quad \sin B = \frac{b \sin C}{c} = \frac{3 \times \frac{\sqrt{3}}{2}}{\sqrt{7}} = \frac{3\sqrt{3}}{2\sqrt{7}}$

Hence, $\sin^2 B = \frac{27}{28}$

39. D

Sol. ${}^{18}C_3 - 3.{}^{7}C_3 = 816 - 105 = 711$

40. D

Sol. Clearly, $f(x) = x + 1 (\text{As } m = 1, c = 1)$

Now, $\tan\left(\tan^{-1}\frac{1}{7} + \tan^{-1}\frac{1}{8} + \tan^{-1}\frac{1}{18}\right)$

$= \tan\left(\tan^{-1}\left(\frac{\frac{1}{7}+\frac{1}{8}}{1-\frac{1}{7}\times\frac{1}{8}}\right) + \tan^{-1}\frac{1}{18}\right)$

$= \tan\left(\tan^{-1}\left(\frac{15}{55}\right) + \tan^{-1}\left(\frac{1}{18}\right)\right) = \tan\left(\tan^{-1}\left(\frac{3}{11}\right) + \tan^{-1}\left(\frac{1}{18}\right)\right)$

$= \tan\left(\tan^{-1}\left(\frac{\frac{3}{11}+\frac{1}{18}}{1-\frac{3}{11}\times\frac{1}{18}}\right)\right) = \tan\left(\tan^{-1}\frac{1}{3}\right) = \frac{1}{3} = f\left(\frac{-2}{3}\right)$

41. C

Sol. $I=\lim\limits_{x\to1}\dfrac{nx^n(x-1)-(x^n-1)}{(e^x-e)\sin\pi x}$

Put $x=1+h$ so that as $x\to1, h\to0$

$\therefore I=-\lim\limits_{h\to0}\dfrac{h.n(1+h)^n-((1+h)^n-1)}{e(e^h-1)\sin\pi h}$

$n.h\left(1+{}^nC_1h+{}^nC_2h^2+{}^nC_3h^3+...\right)$

$I=\lim\limits_{h\to0}\dfrac{-\left(1+{}^nC_1h+{}^nC_2h^2+{}^nC_3h^3+....-1\right)}{\pi e(h^2)\left(\dfrac{e^h-1}{h}\right)\left(\dfrac{\sin\pi h}{\pi h}\right)}$

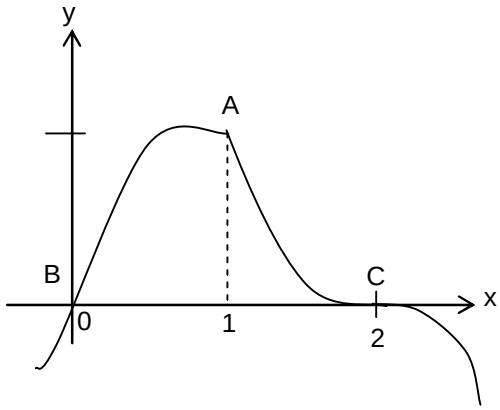
$=-\dfrac{n^2-{}^nC_2}{\pi e}=-\left[\dfrac{2n^2-n(n-1)}{2\pi e}\right]=-\dfrac{n^2+n}{2(\pi e)}=-\dfrac{n(n+1)}{2(\pi e)}$

If $n=100$

$\Rightarrow I=-\left(\dfrac{5050}{\pi e}\right)$

42. C

Sol. A, B, C are the 3 critical points of $y=f(x)$



43. A

Sol. $\dfrac{dy}{dx}=2ax=2x\left(\dfrac{y}{x^2}\right)$

$\dfrac{dy}{dx}=\dfrac{2y}{x}$

Now $m\dfrac{dy}{dx}=-1 \quad\Rightarrow\quad m=-\dfrac{x}{2y} \quad\Rightarrow\quad \dfrac{dy}{dx}=-\dfrac{x}{2y}$

$y^2=-\dfrac{x^2}{2}+c$

44. AC

Sol. $(2\cos\theta)^2=(1-\sin\theta)^2\Rightarrow5\sin^2\theta-2\sin\theta-3=0$

$\Rightarrow \sin\theta = 1$ or $-\frac{3}{5}$, now

If $\sin\theta = 1 \Rightarrow \cos\theta = 0 \Rightarrow E = 3$; If $\sin\theta = -\frac{3}{5}$

Hence $E = \frac{7}{5}$

45. BC

Sol. $I = \frac{1}{2}\int \frac{dt}{t^2+t+1} = \frac{1}{2}\int \frac{dt}{\left(t+\left(\frac{1}{2}\right)\right)^2 + \left(\frac{3}{4}\right)} = \frac{1}{2}.\frac{2}{\sqrt{3}}\tan^{-1}\left(\frac{t+\left(\frac{1}{2}\right)}{\frac{\sqrt{3}}{2}}\right) = \frac{1}{\sqrt{3}}\tan^{-1}\left(\frac{2t+1}{\sqrt{3}}\right)$

$= \frac{1}{\sqrt{3}}\tan^{-1}\left(\frac{2x^2+1}{\sqrt{3}}\right) + C$

46. AB

Sol. Since intercepts are equal in magnitude but opposite in sign.

$\Rightarrow \quad \left.\frac{dy}{dx}\right|_P = 1$

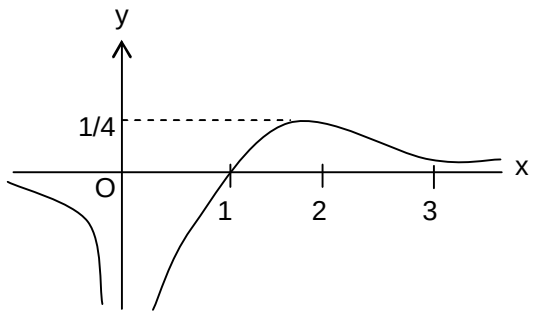
Now $\frac{dy}{dx} = x^2 - 5x + 7 = 1$

$\Rightarrow \quad x^2 - 5x + 6 = 0$

$\Rightarrow \quad x = 2$ or 3

47. BCD

Sol. $f'(x) = \frac{2-x}{x^3}$ and $f''(x) = \frac{x-3}{x^4}$.



48. AB

Sol. $2xy\frac{dy}{dx} = x^2 + y^2 + 1$

Put $y^2 = t$, $x\frac{dt}{dx} = x^2 + t + 1$

$\frac{dt}{dx} - \frac{t}{x} = \frac{x^2+1}{x}$

I.F. $= \frac{1}{x}$

Hence $\frac{t}{x}=\int\frac{x^2+1}{x^2}dx=x-\frac{1}{x}+C$

$\therefore\ y^2=x^2-1,\ C=0$ as $y(1)=0$

Now, $y(x_0)=\sqrt{3}$

$\Rightarrow\ 3=x_0^2-1$

$\Rightarrow\ x_0^2=4$

$\therefore\ x_0=\pm2$

49. BD

Sol. Equation of the line passing through P(1, 4, 3) is: $\frac{x-1}{a}=\frac{y-4}{b}=\frac{z-3}{c}$ .....(i)

Since equation (i) is perpendicular to $\frac{x-1}{2}=\frac{y+3}{1}=\frac{z-2}{4}$ and $\frac{x+2}{3}=\frac{y-4}{2}=\frac{z+1}{-2}$

Hence $2a+b+4c=0$ and $3a+2b-2c=0$

$\therefore\ \frac{a}{-2-8}=\frac{b}{12+4}=\frac{c}{4-3}\ \Rightarrow\ \frac{a}{-10}=\frac{b}{16}=\frac{c}{1}$

Hence the equation of the lines is $\frac{x-1}{-10}=\frac{y-4}{16}=\frac{z-3}{1}$ .....(ii)

Now any point Q on (2) can be taken as $(1-10\lambda,\ 16\lambda+4,\ \lambda+3)$

$\therefore$ Distance of Q from P (1, 4, 3) $=(10\lambda)^2+(16\lambda)^2+\lambda^2=357$

$\Rightarrow\ (100+256+1)\lambda^2=357$

$\Rightarrow\ \lambda=1$ or -1

$\therefore$ Q is $(-9,20,4)$ or $(11,-12,2)$

Hence $a_1+a_2+a_3=15$ or 1

50. ABD

Sol. $\frac{dy}{dx}+\frac{x}{1-x^2}y=\frac{ax}{1-x^2}$

I.F. $e^{\int\frac{x}{1-x^2}dx}=e^{-\frac{1}{2}\log|1-x^2|}=\frac{1}{\sqrt{|1-x^2|}}$

$\frac{y}{\sqrt{|1-x^2|}}=a\int\frac{x}{|1-x^2|^{3/2}}dx+C$

Let $|1-x^2|=v^2$

$-2x\,dx=2v\,dv$

$x\,dx=-v\,dv$

Hence $\frac{y}{v}=-a\int\frac{v\,dv}{v^3}=+\frac{a}{v}+C$

$y=a+Cv$

$y=a+C\sqrt{|1-x^2|}$

$(y-a)^2 = C^2|1-x^2|$

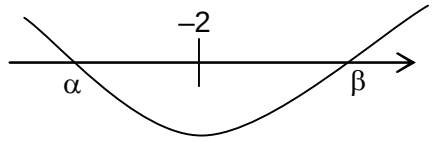
$= C^2(1-x^2)$ or $C^2(x^2-1)$

$(y-a)^2 + C^2x^2 = C^2$ or $(y-a)^2 - C^2x^2 = -C^2$

$\frac{(y-a)^2}{C^2} + \frac{x^2}{1} = 1$ or $\frac{(y-a)^2}{C^2} - \frac{x^2}{1} = -1$ $\Rightarrow$ Centre (0, a)

51. B

Sol. $f(x) = x^2 + (a+2)x + a^2 - a + 2 = 0$



$f(-2) < 0$

$\Rightarrow 4 - 2a - 4 + a^2 - a + 2 < 0$

$\Rightarrow a^2 - 3a + 2 < 0 \quad \Rightarrow 1 < a < 2 \quad \therefore a \in (1, 2)$

Not integral values of a.

52. D

Sol. Given, $x^2 + (a+2)x + a^2 - a + 2 = 0$

Now, $D > 0 \Rightarrow \frac{2}{3} < a < 2$ .....(i)

Product of roots > 0

$\Rightarrow a^2 - a + 2 > 0$, which is true $\forall a \in R$ .....(ii)

and sum of roots > 0

$\Rightarrow -(a+2) > 0$

$\Rightarrow a < -2$ .....(iii)

$\therefore$ Equation $(i) \cap (ii) \cap (iii)$

$\Rightarrow a \in \phi$

53. B

Sol. Required area $= 2 \times \frac{1}{2} \times 4 \times 3 + 2 \times \frac{1}{2} \times 1 \times 3 = 15$

54. A

Sol. $f(-89) - f(-67) + f(46) = f(-1) - f(-3) + f(-2)$

$= 3 - 1 + 2 = 4$

$f(x-2) = f(x+6) \quad \Rightarrow \quad f(x) = f(x+8)$

